

Admissibility variants of Erdős Problem 327: the multipliers $k \geq 2$

Donald Della Pietra

Abstract

For $k \geq 1$ call a set A k -admissible if $a + b \nmid kab$ for all distinct $a, b \in A$, and let $f_k(N)$ be the largest size of a k -admissible subset of $\{1, \dots, N\}$. The case $k = 1$ is the first question of Erdős Problem 327 and $k = 2$ is the second; Sawin proved $f_2(N) \gg N$, and the companion manuscript [3] proved $f_1(N) \geq (1/2 + \varepsilon)N$.

We first settle what the correct benchmark is for each k . A periodic set is k -admissible if and only if every one of its residue classes consists of odd numbers and k is odd. Hence for odd k the odd numbers are k -admissible and are the densest periodic k -admissible set, while for even k no nonempty periodic set is k -admissible. The benchmark is therefore $1/2$ for odd k and 0 for even k ; this is the analogue, for the present problem, of a parity trap, and it changes which target is meaningful.

We then show that the machinery of [3] transfers with k -dependent constants and an unchanged parameter window, provided its dilation 2 is replaced by $\lambda(k)$, the least power of two that is at least k . That replacement is necessary, not cosmetic: with the dilation 2 the construction fails outright whenever $3 \mid k$ or $5 \mid k$, because $\{n, 2n\}$ and $\{2n, 3n\}$ are then never k -admissible. We obtain:

- for every odd k an absolute $\varepsilon_k > 0$ with $f_k(N) \geq (\frac{1}{2} + \varepsilon_k)N$ for large N (Theorem 1.2);
- for every $k \geq 1$ an absolute $c_k > 0$ with $f_k(N) \geq c_k N$ (Theorem 1.3). This is new for even $k \geq 4$; at $k = 2$ it recovers Sawin's theorem, and for odd k it is superseded by the previous item and, indeed, already by the odd numbers.

We do *not* obtain $f_2(N) \geq (1/2 + \varepsilon)N$, the question raised in the Problem 327 discussion thread, and Section 7 isolates the exact obstruction: for even k the method has no dense automatically admissible layer to start from, and the dilation identity shows that no rescaling produces one. All numerical claims are certified by exact rational arithmetic in an accompanying standalone verifier. A Lean formalization is planned and is not part of this manuscript.

1 Introduction

For $k \geq 1$, call a set A of positive integers k -admissible if

$$a + b \nmid kab \quad (a, b \in A, a \neq b), \tag{1}$$

and put

$$f_k(N) = \max\{|A| : A \subseteq \{1, \dots, N\}, A \text{ is } k\text{-admissible}\}.$$

Erdős and Graham [5, p. 37] asked whether a 1-admissible set can be substantially more than the odd numbers, and whether every 2-admissible set has size $o(N)$; these are the two questions of Problem 327 [1]. In unit-fraction language $a + b \mid ab$ says that $1/a + 1/b$ is a unit fraction, and $a + b \mid 2ab$ says that the average of $1/a$ and $1/b$ is a unit fraction.

The state of the art we start from is the following.

- Sawin [7] proved $f_2(N) \gg N$, answering the second question negatively. The constant is not made explicit.
- The companion manuscript [3] proved $f_1(N) \geq (1/2 + \varepsilon)N$ for an absolute $\varepsilon > 0$.
- In the discussion thread [4], leon2k2k2k obtained the upper bounds $f_1(N) \leq (0.7769 + o(1))N$, $f_2(N) \leq (0.7630 + o(1))N$ and $f_3(N) \leq (0.6089 + o(1))N$ [6], and asked whether $f_2(N) \geq (1/2 + o(1))N$.

For odd k the odd numbers give the trivial bound $f_k(N) \geq N/2 + O(1)$, as we recall in Section 2. Beyond that, no lower bound for f_k with $k \geq 3$ — in particular none at all for even $k \geq 4$, where the trivial bound is unavailable — appears in the literature or in the thread; a dated search is reported in Section 9.

1.1 The correct benchmark

Before asking whether one can beat density $1/2$ it is necessary to know whether $1/2$ is available at all. It is not, for even k . The following is proved in Section 2; it is sharp, and it is what makes the odd and even cases genuinely different problems.

Theorem 1.1 (periodic benchmark). *Let $k, M \geq 1$ and $0 \leq r < M$, and let $C_{r,M} = \{n \geq 1 : n \equiv r \pmod{M}\}$.*

1. *If k is odd, M is even and r is odd, then $C_{r,M}$ is k -admissible.*
2. *In every other case $C_{r,M}$ contains distinct a, b with $a + b \mid kab$; an explicit such pair is displayed in (2).*

Consequently, a union of residue classes to a common modulus is k -admissible if and only if each of its classes is of type (1). For odd k every periodic k -admissible set consists of odd numbers and so has density at most $1/2$, with equality for the odd numbers; for even k the only periodic k -admissible set is the empty set.

So the benchmark density is $1/2$ for odd k and 0 for even k , and “beating the odd numbers” is a meaningful goal exactly when k is odd. For even k even a single residue class must be pruned, and every construction has to be aperiodic from the outset.

1.2 Results

Theorem 1.2 (odd multipliers). *For every odd $k \geq 1$ there are constants $\varepsilon_k > 0$ and N_k such that*

$$f_k(N) \geq \left(\frac{1}{2} + \varepsilon_k\right)N \quad (N \geq N_k).$$

For $k = 1$ this is [3, Theorem 1.1] and contains nothing new; the content here is that the argument transfers to every odd k with the *same* certified parameter window and only k -dependent multiplicative constants. Combined with the upper bound of [6] — itself a thread posting, independently checked but not refereed — this would give, for example,

$$\frac{1}{2} < \liminf_{N \rightarrow \infty} \frac{f_3(N)}{N} \leq \limsup_{N \rightarrow \infty} \frac{f_3(N)}{N} \leq 0.6089.$$

Theorem 1.3 (all multipliers). *For every $k \geq 1$ there is a constant $c_k > 0$ such that $f_k(N) \geq c_k N$ for all sufficiently large N .*

For odd k , Theorem 1.3 is trivial: the odd numbers already have density $1/2$. Its content is for even k , where by Theorem 1.1 no such trivial set exists. At $k = 2$ it reproves Sawin's theorem by the route of [3, §4]; for even $k \geq 4$ it is, to our knowledge, the first lower bound of any kind. It uses only the source half of the machinery, not the two-endpoint estimate, and is therefore the more robust of the two results. Because both analytic inputs carry unspecified absolute constants, c_k is not effective as stated; see Section 8.

1.3 What does not close

We do not prove $f_2(N) \geq (1/2 + \varepsilon)N$, nor any bound for even k beyond positive density. Section 7 records three separate reasons, each of which is proved rather than asserted: the absence of a periodic layer (Theorem 1.1), the dilation identity (Proposition 2.4), which shows that every rescaled layer of an even- k construction is again an even-multiplier problem, and the fact that the roughness cutoff, which is what buys the saving in the source estimate, caps the density of the source at $O(\rho_L)$. We also record, in Proposition 7.1, a corrected form of the largest-prime-factor observation from the thread, and show that its contraction constant $\log 2 < 1$ prevents it from bootstrapping any density.

1.4 Method and organisation

Section 2 is self-contained and elementary. Section 3 recalls the two analytic inputs and the centered anatomy of [3] verbatim; nothing there is new. Section 4 generalises the source estimate from the modulus 2 to an arbitrary fixed modulus m and proves Theorem 1.3. Section 5 generalises the two-endpoint mixed estimate to odd k , and this is where the dilation $\lambda(k)$ of (12) is introduced and where Remark 5.1 explains why the naive dilation 2 is fatal. Section 6 assembles Theorem 1.2. Section 7 treats the even case and the obstruction. Section 8 lists every step that is imported, flagged, or unverified, and Section 9 describes the verifier, the prior-work search and the AI-use disclosure.

2 The correct benchmark

Everything in this section is elementary and is confirmed exhaustively by the verifier.

Lemma 2.1 (conflict criterion). *Let $a \neq b$ be positive integers, $g = (a, b)$, and $x = (a + b)/g$. Then for every $k \geq 1$,*

$$a + b \mid kab \iff \frac{x}{(x, k)} \mid g.$$

Proof. Write $a = gu$, $b = gv$ with $(u, v) = 1$, so $x = u + v$ and $a + b = gx$, $kab = kg^2uv$. Thus $a + b \mid kab$ if and only if $x \mid kguv$. Now $(x, u) = (u + v, u) = (v, u) = 1$ and likewise $(x, v) = 1$, so $(x, uv) = 1$ and the condition is $x \mid kg$. Finally, for every prime p ,

$$v_p(x) \leq v_p(k) + v_p(g) \iff v_p(g) \geq v_p(x) - \min(v_p(x), v_p(k)) = v_p\left(\frac{x}{(x, k)}\right),$$

which is the stated divisibility. □

Corollary 2.2 (two-adic restriction). *Let a, b be distinct odd integers with $a + b \mid kab$. Then $v_2(a + b) \leq v_2(k)$. In particular, for $k = 2$ one has $a \equiv b \pmod{4}$: the conflict graph of the multiplier 2 on the odd numbers has no edge between the classes 1 and 3 modulo 4.*

Proof. Here $g \mid a$ is odd, so $v_2(x) = v_2(gx) = v_2(a + b)$. By Lemma 2.1 we have $x \mid kg$, whence $v_2(x) \leq v_2(k)$. For $k = 2$ this gives $v_2(a + b) \leq 1$, and since a, b are odd, $a + b \equiv 2 \pmod{4}$, i.e. $a \equiv b \pmod{4}$. $\square$

Proof of Theorem 1.1. (1) If a, b are odd then $a + b$ is even while kab is odd, so (1) holds. A class $C_{r,M}$ with M even and r odd consists of odd numbers.

(2) Suppose we are not in case (1), i.e. k is even, or M is odd, or r is even. Put

$$u = 1, \quad v = 1 + 2M, \quad x = u + v = 2(1 + M), \quad y = \frac{x}{(x, k)}.$$

We claim $(y, M) \mid r$. Let p be a prime dividing M . If p is odd then $x = 2 + 2M \equiv 2 \pmod{p}$, so $p \nmid x$ and hence $p \nmid y$. If $p = 2$, then M is even, so $1 + M$ is odd and $v_2(x) = 1$; therefore $v_2(y) = 1 - \min(1, v_2(k))$, which is 0 when k is even and 1 when k is odd. Consequently

$$(y, M) = \begin{cases} 2, & M \text{ even and } k \text{ odd,} \\ 1, & \text{otherwise,} \end{cases}$$

and in each of the three cases excluded from (1) we get $(y, M) \mid r$: if k is even or M is odd then $(y, M) = 1$, and if M is even, k is odd and r is even then $(y, M) = 2 \mid r$.

Since $(y, M) \mid r$, the simultaneous congruences $g \equiv 0 \pmod{y}$ and $g \equiv r \pmod{M}$ have a positive solution g . Put

$$a = g, \quad b = g(1 + 2M). \quad (2)$$

Then $a \equiv r$ and $b \equiv g \equiv r \pmod{M}$, and $a \neq b$ because $1 + 2M > 1$. Moreover $(a, b) = g$ and $(a + b)/g = 1 + v = x$, so Lemma 2.1 applies with this g and this x : since $y \mid g$ we conclude $a + b \mid kab$.

For the final statement, a union A of classes modulo M is k -admissible only if each individual class is, since a conflict inside one class is a conflict inside A ; and conversely, if every class of A is of type (1) then A consists of odd numbers and is k -admissible by (1). For odd k the admissible classes are exactly the odd classes, whose union has density at most $1/2$; for even k case (1) is vacuous. $\square$

Remark 2.3. The threshold in Theorem 1.1 is delicate. The class 7 mod 10 contains no 2-conflict among its elements below 897; its first three conflicts are (117, 897), (357, 1377) and (77, 1617). A purely computational search in a box of size a few hundred would have reported the wrong benchmark. Note also that the witness (2) is explicit but not minimal: at $k = 2$, $M = 10$, $r = 7$ it has $y = 11$ and $g = 77$, returning the third of those pairs and not the first. The construction is designed to be uniform in (k, M, r) , not sharp.

Proposition 2.4 (dilation identity). *Let λ be a positive integer. A set A satisfies $\lambda a + \lambda b \nmid k(\lambda a)(\lambda b)$ for all distinct $a, b \in A$ if and only if A is $k\lambda$ -admissible. Consequently*

$$f_k(N) \geq f_{k\lambda}(\lfloor N/\lambda \rfloor).$$

Proof. $\lambda(a + b) \mid k\lambda^2 ab$ if and only if $a + b \mid k\lambda ab$. $\square$

This is the identity behind the two-layer construction: the even part of a 1-admissible set is 2 times a 2-admissible set, an equivalence already recorded in the thread. It also shows that for even k every dilated layer carries an even multiplier, so Theorem 1.1 denies a periodic layer at every scale.

3 Notation and imported analytic inputs

All logarithms are natural, $\Omega(n)$ is the number of prime factors of n with multiplicity, $P^-(n)$ is the least prime factor of n with $P^-(1) = +\infty$, and for real $L > 2$

$$\Omega_L(n, x) = \sum_{\substack{p^\nu \parallel n \\ L \leq p \leq x}} \nu, \quad \rho_L = \prod_{p < L} \left(1 - \frac{1}{p}\right).$$

An integer is L -rough if it has no prime factor below L , and (L, A, K) -regular if

$$\Omega_L(n, x) \leq A \log \frac{\log x}{\log L} + K \quad (x \geq L). \quad (3)$$

For an integer n and a cutoff L we write $n = n_{<L} n_{\geq L}$ for the factorisation into the L -smooth and L -rough parts.

We import three statements from [3] without change. The first two are the analytic inputs; the third is the centered tail estimate, whose proof involves no multiplier and is reproduced there in full.

Proposition 3.1 (one-variable mean value; [3, Prop. 2.1], after [8, Thm. III.3.5]). *Let g be a nonnegative multiplicative function with $\sum_{p \leq y} g(p) \log p \leq C_1 y$ for all $y \geq 1$ and $\sum_p \sum_{\nu \geq 2} g(p^\nu) \log(p^\nu) p^{-\nu} \leq C_2$. Then*

$$\sum_{n \leq Y} g(n) \ll (C_1 + C_2 + 1)^Y \prod_{p \leq Y} \left(1 - \frac{1}{p} + \sum_{\nu \geq 1} \frac{g(p^\nu)}{p^\nu}\right)$$

with an absolute implied constant.

Proposition 3.2 (three linear forms; [3, §2], after [2, Thm. 3.1]). *Let $Q_1, Q_2, Q_3 \in \mathbb{Z}[U, V]$ be fixed, primitive, pairwise nonproportional linear forms, and let $F : \mathbb{N}^3 \rightarrow \mathbb{R}_{\geq 0}$ be multiplicative in the sense that $F(a_1 b_1, a_2 b_2, a_3 b_3) \leq F(b_1, b_2, b_3)$ whenever $(a_1 a_2 a_3, b_1 b_2 b_3) = 1$. Then, on every box of side lengths comparable to X ,*

$$\sum_{(u,v)} F(Q_1(u, v), Q_2(u, v), Q_3(u, v)) \ll X^2 E_{F,Q}(CX) \prod_{3 < p \leq CX} \left(1 - \frac{\rho_Q^+(p)}{p^2}\right), \quad (4)$$

where $\rho_Q^+(p) = \#\{(u, v) \bmod p : p \mid Q_1 Q_2 Q_3(u, v)\}$, C depends only on the forms and the box aspect ratio, and $E_{F,Q}$ is bounded by the product of its local prime-power factors. For the triples used below a local pattern with exactly one positive valuation e has density

$$\frac{(1 - p^{-1})^2}{p^e} \quad (5)$$

at every odd prime, and patterns with two or three positive valuations vanish. The implied constant is uniform over the class once the forms and the box aspect ratio are fixed; this is what permits F to depend on the cutoff L .

Lemma 3.3 (centered maximal tail; [3, Lem. 3.1]). *Fix $1 < z < 2$ and $A > 0$ with $A \log z > z - 1$. Uniformly for real $L > 2$, $Y \geq 1$ and $K \geq 0$,*

$$\#\{n \leq Y : n \text{ is not } (L, A, K)\text{-regular}\} \ll_{A,z} Y z^{-K},$$

and among L -rough integers the same count is $\ll_{A,z} Y \rho_L z^{-K}$.

4 The source estimate for an arbitrary modulus

Fix an integer $m \geq 1$. This section proves the existence of an m -admissible set of positive density. The argument is [3, §4] with 2 replaced by m ; we display the four places where the multiplier intervenes, each of which costs only a constant.

Throughout, $\delta \in (0, 1)$, $1 < A < 3/(2 \log 4)$, $K \geq 0$ and $L > m$ are fixed, and

$$S = S(N; m, L, A, K) = \{b \in [\delta N, N] : P^-(b) \geq L, \text{ } b \text{ is } (L, A, K)\text{-regular}\}.$$

Lemma 4.1 (source density; [3, Lem. 4.1]). *There is $K_0 = K_0(A, \delta)$, independent of L and of m , such that for $K \geq K_0$, every fixed $L > 2$ and all large N , $|S| \geq \frac{1-\delta}{2} N \rho_L$.*

Call $b \in S$ *bad* if there is $c \leq N$, $c \neq b$, with

$$b + c \mid mbc, \quad \Omega(c) \leq \Omega(b), \quad (6)$$

and let $T = T(N; m, L, A, K)$ be the number of bad b .

Lemma 4.2 (generalised Sawin coordinates). *Let $b \in S$ and c satisfy (6). Put $g = (b, c)$, $u = b/g$, $v = c/g$ and $x = u + v$. Then*

$$(u, v) = 1, \quad ux \mid mb, \quad v \leq \delta^{-1}u, \quad \Omega(v) \leq \Omega(u), \quad u \geq L, \quad x > L, \quad (7)$$

and, writing $mb = uxd$, the L -smooth part of uxd is exactly m ; in particular u is L -rough and $x_{<L}d_{<L} = m$.

Proof. By Lemma 2.1, $x \mid mg$, hence $ux \mid mgu = mb$; the remaining inequalities in (7) follow from $c/b = v/u$, the ranges of b, c , and complete additivity of Ω . Since $u \mid b$ and b is L -rough, u is L -rough. If $u = 1$ then $\Omega(v) \leq \Omega(u) = 0$ forces $v = 1$ and $c = b$, which is excluded; therefore $u \geq L$ and $x = u + v > L$. Finally $uxd = mb$ with b L -rough and m itself L -smooth, so the L -smooth part of the left side is m . $\square$

Compared with [3, (4.3)] the exclusion $x \geq L$ is obtained here from the roughness of u rather than from a case analysis at 2; this is the form that survives an arbitrary multiplier.

Proposition 4.3 (source estimate). *Put $M = A \log 4$. For every fixed K, L and all sufficiently large N ,*

$$\frac{T(N; m, L, A, K)}{N \rho_L} \ll_{A, \delta, m} 4^K \left\{ (\log L)^{-3/2} + (\log L)^{1/4-M} (\log N)^{M-3/2} \right\}. \quad (8)$$

Proof. We follow [3, Prop. 4.2], indicating the four changes.

(i) *The indicator.* Since $L > m$, every prime factor of m is below L , so $\Omega_L(b, x) = \Omega_L(mb, x)$. At the scale x , Lemma 4.2 and $mb = uxd$ give

$$\Omega_L(b, x) = \Omega_L(u, x) + \Omega_L(x, x) + \Omega_L(d, x) = \Omega(u) + \Omega_L(x, x) + \Omega_L(d, x),$$

where the last step used $u \leq x$ and the roughness of u . Since $x_{<L} \mid m$, $\Omega_L(x, x) \geq \Omega(x) - \Omega(m)$, so regularity at x yields

$$\Omega(u) + \Omega(x) + \Omega_L(d, x) \leq A \log \frac{\log x}{\log L} + K + \Omega(m).$$

Exponentiating with base 4 bounds the indicator by

$$4^{K+\Omega(m)} \left(\frac{\log x}{\log L} \right)^M 4^{-\Omega(u)} 4^{-\Omega(x)} 4^{-\Omega_L(d,x)}. \quad (9)$$

Relative to the case $m = 2$ this costs the constant $4^{\Omega(m)-1}$.

(ii) *The residual sum.* Write $d = d_{<L} d_{\geq L}$ with $d_{<L} \mid m$. Since Ω_L ignores primes below L ,

$$R(D) = \sum_{\substack{d \leq D \\ d_{<L} \mid m}} 4^{-\Omega_L(d,x)} = \sum_{d_0 \mid m} \sum_{\substack{d_1 \leq D/d_0 \\ P^-(d_1) \geq L}} 4^{-\Omega_L(d_1,x)},$$

a sum of $\tau(m)$ copies of the sum treated in [3, (4.6)]. Proposition 3.1 applies with local factors $1 - p^{-1}$ for $p < L$, $1 - \frac{3}{4p} + O(p^{-2})$ for $L \leq p \leq x$, and $1 + O(p^{-2})$ elsewhere, giving

$$R(D) \ll_m D \left\{ (\log x)^{-3/4} (\log L)^{-1/4} + (\log(3D))^{-3/4} \right\}. \quad (10)$$

(iii) *The three-form sum.* Because $\Omega(v) \leq \Omega(u)$,

$$4^{-\Omega(u)} 4^{-\Omega(x)} \leq g_{1,L}(u) g_2(v) g_{3,L}(x),$$

where $g_{1,L}(n) = 2^{-\Omega(n)} \mathbf{1}_{P^-(n) \geq L}$, $g_2(n) = 2^{-\Omega(n)}$ and $g_{3,L}(n) = 4^{-\Omega(n)} \mathbf{1}_{n_{<L} \mid m}$. Setting $F_L(n_1, n_2, n_3) = \mathbf{1}_{(n_1, n_2) = 1} g_{1,L}(n_1) g_2(n_2) g_{3,L}(n_3)$, the hypothesis of Proposition 3.2 holds uniformly in L , and we apply it to $Q = (U, V, U + V)$, for which $\rho_Q^+(p) = 3p - 2$ at every prime. By (5) the local factor at an odd $p \geq L$ is exactly

$$1 + (1 - p^{-1})^2 \left(\frac{1}{2p-1} + \frac{1}{2p-1} + \frac{1}{4p-1} \right) = 1 + \frac{5}{4p} + O(p^{-2}),$$

while at $2 < p < L$ with $p \nmid m$ only the V -axis survives, giving $1 + \frac{1}{2p} + O(p^{-2})$. At the finitely many primes dividing $2m$ the $U + V$ -axis is permitted up to valuation $v_p(m)$ and the factor is multiplied by a bounded amount, absorbed into the constant. Mertens' estimate and the root-sieve product $\ll (\log X)^{-3}$ give, on $u \in [X, 2X]$,

$$\sum_{\substack{X \leq u < 2X \\ 0 < v \leq \delta^{-1}u \\ (u,v)=1}} g_{1,L}(u) g_2(v) g_{3,L}(u+v) \ll_{\delta,m} X^2 (\log X)^{-7/4} (\log L)^{-3/4}, \quad (11)$$

with every nonzero block satisfying $X \gg_{\delta} L$.

(iv) *The dyadic assembly.* This is identical to [3, Prop. 4.2], with $D = mN/(ux)$ in (10); the replacement of m for 2 changes D by the bounded factor $m/2$, hence changes $\log(3D)$ by $O_m(1)$. Combining (9), (10) and (11) and dividing by $N\rho_L$, one block contributes $\ll 4^K (\log L)^{-M} (\log X)^{M-5/2}$ from the first residual term; since $M < 3/2$ the dyadic sum over $X \gg L$ converges to $\ll (\log L)^{M-3/2}$, leaving $4^K (\log L)^{-3/2}$. The second residual term contributes the second summand of (8) exactly as there. $\square$

Proof of Theorem 1.3. For odd k the odd numbers already give $f_k(N) \geq \lceil N/2 \rceil$ by Theorem 1.1(1), with a far better constant than the one below; so assume k is even. Apply the above with $m = k$, $\delta = 1/2$ and a fixed $A \in (1, 3/(2 \log 4))$. (The argument below in fact works verbatim for every $k \geq 2$; only for even k does it say anything.) Choose $K \geq K_0(A, \delta)$ from Lemma 4.1. Because $M = A \log 4 < 3/2$, the first term of the right side of (8) can be made at most $\frac{1-\delta}{8}$ by fixing $L > k$

large enough, and with that L fixed the second term tends to 0, so is also at most $\frac{1-\delta}{8}$ for large N . Hence $T \leq \frac{1-\delta}{4} N \rho_L \leq \frac{1}{2} |S|$ by Lemma 4.1. Delete every bad b from S and call the survivors B . If distinct $b, d \in B$ satisfied $b + d \mid mbd$, then relabelling so that $\Omega(d) \leq \Omega(b)$ makes b bad, a contradiction; hence B is k -admissible. For all large N , $|B| \geq \frac{1}{2} |S| \geq \frac{1-\delta}{4} N \rho_L = \frac{1}{8} N \rho_L$, so $c_k = \rho_L/8$ works. $\square$

5 The two-endpoint estimate for odd k

From now on k is odd, and we fix the dilation

$$\lambda = \lambda(k) = \min\{2^j : j \geq 1, 2^j \geq k\}, \quad (12)$$

the least even power of two that is at least k . The construction of Theorem 1.2 is

$$A_N = (O_N \setminus \Gamma_N) \cup \lambda B_N \subseteq \{1, \dots, \lambda N\},$$

where O_N is a set of odd integers up to λN , $B_N \subseteq [\delta N, N]$ is $k\lambda$ -admissible — so that λB_N has no internal conflict, by Proposition 2.4 — and Γ_N removes the mixed conflicts. Two odd elements never conflict because k is odd. It therefore remains to count the pairs (a, b) with a odd, $b \in B_N$ and

$$a + \lambda b \mid k\lambda ab. \quad (13)$$

The choice of λ is not cosmetic. For $k = 1$ one may take $\lambda = 2$, as [3] does, but that choice is fatal as soon as $3 \mid k$ or $5 \mid k$.

Remark 5.1 (why $\lambda = 2$ fails). Let n be odd. If $3 \mid k$ then $n + 2n = 3n$ divides $k \cdot n \cdot 2n = 2kn^2$, so $\{n, 2n\}$ is never k -admissible. With $\lambda = 2$ each $b \in B_N$ would then force the deletion of the odd number b itself from O_N , and that loss cancels the entire gain $|B_N|$: the construction yields nothing. If $5 \mid k$ then $3n + 2n = 5n$ divides $k \cdot 3n \cdot 2n$, so $\{2n, 3n\}$ is never k -admissible, and the same objection applies to a positive proportion of B_N . In the coordinates of Lemma 5.2 these are exactly the families with $u = 1$, which exist precisely when $w + \lambda$ divides k for some admissible w . Choosing $\lambda \geq k$ forces $w + \lambda > k$ and eliminates all of them; this is the sole purpose of (12). These families are invisible at $k = 1$, which is why they do not arise in [3].

Lemma 5.2 (mixed coordinates). *Let k be odd, let λ be as in (12), let a be odd and let b be odd and L -rough, and suppose (13) holds. Put $g = (a, b)$, $w = a/g$, $u = b/g$, $x = w + \lambda u$, $\kappa = (x, k)$ and $x' = x/\kappa$. Then $(u, w) = 1$, all of $t = g/x'$, u , w , x are odd, $x' \mid g$,*

$$a = tx'w, \quad b = tx'u, \quad (14)$$

and κ is the L -smooth part of x , so that $\kappa \mid k$ and t, u, x' are L -rough. Conversely every such tuple gives a conflict.

If in addition $a \leq \lambda N$, $b \geq \delta N$ and $L > \lambda(1 + \delta^{-1})$, then

$$u \geq L, \quad \text{and hence} \quad x > \lambda u \geq \lambda L. \quad (15)$$

Proof. Apply Lemma 2.1 to the pair $(a, \lambda b)$ with multiplier $k\lambda$. Here $(a, \lambda b) = (a, b) = g$ because a is odd and λ is a power of two; the reduced pair is $(w, \lambda u)$, coprime because w is odd and $(w, u) = 1$; and $(a + \lambda b)/g = w + \lambda u = x$, which is odd. Lemma 2.1 makes the conflict equivalent to $x/(x, k\lambda) \mid g$, and x is odd while λ is a power of two, so $(x, k\lambda) = (x, k) = \kappa$. Hence the conflict is equivalent to x' dividing g , which gives (14) with $t = g/x'$. Since $b = tx'u$ is L -rough, so are t , x' and u ; therefore

the L -smooth part of $x = \kappa x'$ is κ , and $\kappa \mid k$ by definition. All the variables are odd because a and b are and k is. The converse is immediate from Lemma 2.1.

For (15), suppose $u = 1$. Then $g = b$, so $w = a/b \leq \lambda N/(\delta N) = \lambda \delta^{-1}$ and

$$x = w + \lambda \leq \lambda(1 + \delta^{-1}) < L.$$

Since $x' \mid g = b$ and b is L -rough, x' is L -rough; as $1 \leq x' \leq x < L$, this forces $x' = 1$, that is $x \mid k$. But $x = w + \lambda > \lambda \geq k$ by (12), and no divisor of k exceeds k . This contradiction shows $u \neq 1$. Since $u \mid b$ and b is L -rough, u is L -rough, so $u \geq L$; and $x = w + \lambda u > \lambda u \geq \lambda L$. $\square$

Fix positive A_b, A_o, K_b, K_o , bases $q_b, q_o > 1$, $\delta \in (0, 1)$, λ as in (12), and $L > \lambda(1 + \delta^{-1})$. Let $S \subseteq [\delta N, N]$ be the L -rough integers obeying the centered (A_b, K_b) bound, and let O be the odd $a \leq \lambda N$ obeying the centered (A_o, K_o) bound. Put

$$E_{\text{mix}}^{(k)}(N) = \#\{(a, b) \in O \times S : a + \lambda b \mid k \lambda a b\},$$

$$\alpha = q_b^{-1}, \quad \beta = q_o^{-1}, \quad s = (q_b q_o)^{-1}, \quad M = A_b \log q_b + A_o \log q_o, \quad E = M + \alpha + \beta + 2s - 4.$$

Proposition 5.3 (two-endpoint estimate). *If $E < -1$, then*

$$E_{\text{mix}}^{(k)}(N) \ll_{k, \delta} q_b^{K_b} q_o^{K_o} \left\{ N(\log L)^{-2} + o_L(N) \right\}, \quad (16)$$

where the normalised o_L -function may be chosen independently of K_b, K_o .

Proof. We follow [3, Prop. 5.2]. Sum over the at most $\tau(k)$ values of $\kappa \mid k$ separately; fix one.

Anatomy. By Lemma 5.2, at the scale x ,

$$\Omega_L(b, x) = \Omega_L(t, x) + \Omega(u) + \Omega(x) - \Omega(\kappa),$$

$$\Omega_L(a, x) = \Omega_L(t, x) + \Omega(x) - \Omega(\kappa) + \Omega_L(w, x).$$

Here we used $u < x$, $x' \leq x$ and the roughness of t, u, x' . We also use $\Omega_L(w, x) \geq \Omega_{\geq L}(w) - 1$: indeed $w \leq \lambda \delta^{-1} u < \lambda \delta^{-1} x$, so once $x \geq \lambda \delta^{-1}$ — guaranteed by (15) — the integer w has at most one prime factor exceeding x , since two would force $w > x^2$. (The corresponding step of [3] is written as an equality; the discrepancy costs a single factor q_o and is flagged in Section 8.) Exponentiating the two regularity bounds with bases q_b, q_o majorises the joint indicator by

$$C_k q_b^{K_b} q_o^{K_o} \left(\frac{\log x}{\log L} \right)^M \alpha^{\Omega(u)} \beta^{\Omega_{\geq L}(w)} s^{\Omega(x)} s^{\Omega_L(t, x)}, \quad C_k = (q_b q_o)^{\Omega(k)} q_o. \quad (17)$$

This is [3, (5.4)] times the constant C_k .

Residual sum. With $Y = \kappa N/(ux) \asymp_k N/(ux)$, Proposition 3.1 applied to $h_{L, x}(t) = \mathbf{1}_{P^-(t) \geq L} s^{\Omega_L(t, x)}$ gives

$$\sum_{\substack{\delta Y \leq t \leq Y \\ P^-(t) \geq L}} s^{\Omega_L(t, x)} \ll_s Y \left\{ (\log x)^{s-1} (\log L)^{-s} + (\log(3Y))^{s-1} \right\}, \quad (18)$$

exactly as in [3, (5.5)]; the bounded factor κ changes Y and $\log(3Y)$ by $O_k(1)$.

Three-form sum. The forms are $Q = (U, W, \lambda U + W)$. Because λ is a power of two it is invertible modulo every odd prime p , so the three root lines are distinct there and $\rho_Q^+(p) = 3p - 2$, exactly as for $\lambda = 2$; all the local computations below are therefore independent of λ . At $p = 2$ all three values

are forced odd and the local factor is 1. For $2 < p < L$ with $p \nmid k$ only the W -axis is permitted and (5) gives the exact factor

$$\mathcal{E}_p = 1 + (1 - p^{-1})^2 \sum_{e \geq 1} p^{-e} = 1 + \frac{1}{p} - \frac{1}{p^2}; \quad (19)$$

for the finitely many $p \mid k$ with $p < L$ the $\lambda U + W$ -axis is permitted up to valuation $v_p(\kappa) \leq v_p(k)$, which multiplies (19) by a factor between 1 and $3/2$ (certified in the supplement), contributing an $O_k(1)$ constant. For $p \geq L$ the complete factor is

$$\mathcal{E}_p = 1 + (1 - p^{-1})^2 \left(\frac{\alpha}{p - \alpha} + \frac{\beta}{p - \beta} + \frac{s}{p - s} \right) = 1 + \frac{\alpha + \beta + s}{p} + O_{\alpha, \beta, s}(p^{-2}), \quad (20)$$

with an error constant independent of L . Hence, on $u \in [X, 2X]$,

$$\sum_{\substack{X \leq u < 2X \\ 0 < w \leq \lambda \delta^{-1} u \\ (u, w) = 1}} \alpha^{\Omega(u)} \mathbf{1}_{2 \nmid w} \beta^{\Omega_{\geq L}(w)} s^{\Omega(\lambda u + w)} \mathbf{1}_{P^-(u) \geq L, (\lambda u + w) < L \mid k} \ll_k X^2 (\log X)^{\alpha + \beta + s - 3} (\log L)^{1 - \alpha - \beta - s}, \quad (21)$$

with $ux \asymp_{\delta, k} X^2$. Every block with a nonzero summand has $X \geq L$: this is exactly (15), and it is the point at which the choice $\lambda \geq k$ is used. Without it the blocks with $u = 1$ would survive, and by Remark 5.1 they carry a positive proportion of the source whenever $3 \mid k$ or $5 \mid k$, destroying the estimate.

Assembly. Identical to [3, Prop. 5.2]. Inserting the first term of (18) into (17) and using (21), one block contributes $\ll C_k q_b^{K_b} q_o^{K_o} N (\log X)^E (\log L)^{-E-3}$; since $E < -1$ the dyadic sum from $X \geq L$ is $\ll (\log L)^{E+1}$, leaving exactly $N (\log L)^{-2}$. The second residual term is $o_L(N)$ after normalisation by the same three-way split at $\log X = \frac{1}{4} \log N$ as there. Summing over the $\tau(k)$ values of κ multiplies the result by $O_k(1)$. $\square$

6 The parameter window and the proof of Theorem 1.2

The window is that of [3, §6]; no inequality in it mentions k .

Lemma 6.1 (certified window; [3, §6]). *The rationals*

$$\begin{aligned} A_b &= 1.000001, & q_b &= 2.48933, & A_o &= 1.16312, \\ q_o &= 1.34288, & z_o &= 1.34305, & c &= 3.3912 \end{aligned} \quad (22)$$

satisfy the five strict inequalities

$$A_b \log 4 < \frac{3}{2}, \quad A_o \log z_o > z_o - 1, \quad c \log z_o > 1, \quad c \log q_o < 1, \quad E < -1, \quad (23)$$

with the certified one-sided margins of Table 1.

Theorem 6.2. *Let k be odd and let $\lambda = \lambda(k)$ be as in (12). There is an absolute $\eta_k > 0$ such that for every sufficiently large N there is a k -admissible $A_N \subseteq \{1, \dots, \lambda N\}$ with*

$$|A_N| \geq \frac{\lambda N}{2} + \eta_k N.$$

quantity	certified one-sided margin
$A_b \log 4 - 5/2 < -1$	upper value < -1.1137042525
$A_o \log z_o - (z_o - 1) > 0$	> 0.0000042730
$c \log z_o - 1 > 0$	> 0.0002111998
$1 - c \log q_o > 0$	> 0.0002180771
$E < -1$	upper value < -1.0004077237

Table 1: Strict parameter margins, recertified by rational interval arithmetic in the supplement. Every logarithm is enclosed in a rational interval of width below 10^{-80} .

Proof. Fix (22) and $\delta = 1/2$. Choose $z_b \in (1, 2)$ with $A_b \log z_b > z_b - 1$, and then, by Lemma 4.1, a fixed K_b retaining a uniform positive proportion of every L -rough source interval. This choice precedes L .

For each N let $S_N \subseteq [N/2, N]$ be the L -rough integers satisfying the centered (A_b, K_b) condition, so that $|S_N| \geq \frac{1}{4}N\rho_L$ for fixed L and all large N . Delete from S_N every b admitting $d \leq N$, $d \neq b$, with $b + d \mid k\lambda bd$ and $\Omega(d) \leq \Omega(b)$, and call the survivors B_N . By the argument in the proof of Theorem 1.3, B_N is $k\lambda$ -admissible. Applying Proposition 4.3 with $m = k\lambda$ and using $A_b \log 4 < 3/2$, choose $L > \max\{k\lambda, \lambda(1 + \delta^{-1})\} = k\lambda \vee 3\lambda$, after K_b , so large that the first term of the right side of (8) is at most $\frac{1}{8}$; the second term is then at most $\frac{1}{8}$ for large N , so the total loss is at most $\frac{1}{4}N\rho_L \leq \frac{1}{2}|S_N|$ and

$$|B_N| \geq \frac{1}{2}|S_N| \geq \frac{1}{8}N\rho_L \quad (24)$$

for all large N .

Put $K_o = \lceil c \log \log L \rceil$ and let O_N be the odd $a \leq \lambda N$ satisfying the centered (A_o, K_o) condition, so that the total number of odd $a \leq \lambda N$ is $\frac{\lambda N}{2} + O(1)$. By Lemma 3.3 and $c \log z_o > 1$,

$$\frac{\lambda N}{2} - |O_N| \ll_{\lambda} N z_o^{-K_o} \ll_{\lambda} N (\log L)^{-c \log z_o} = o_L(N\rho_L),$$

so enlarging L gives

$$\frac{\lambda N}{2} - |O_N| \leq \frac{1}{32}N\rho_L. \quad (25)$$

Let $E_N = E_{\text{mix}}^{(k)}(N)$ be the number of $(a, b) \in O_N \times S_N$ with $a + \lambda b \mid k\lambda ab$. By Proposition 5.3 and $E < -1$,

$$E_N \ll_k q_b^{K_b} q_o^{K_o} \left\{ N(\log L)^{-2} + o_L(N) \right\}.$$

Since K_b is fixed and $q_o^{K_o} \ll (\log L)^{c \log q_o}$ with $c \log q_o < 1$, the first term is $\ll_k N(\log L)^{-2+c \log q_o} = o_L(N\rho_L)$, because $\rho_L \asymp 1/\log L$ and $-2 + c \log q_o < -1$. The k -dependent constants C_k and $\tau(k)$, and the dilation $\lambda = \lambda(k)$, are all fixed before L , so a further enlargement of L , depending on k , absorbs them. After L and hence K_o are fixed, take N large enough for the rough source asymptotic, the second source residual range, the odd-host bound and the normalised mixed o_L -term. Thus

$$E_N \leq \frac{1}{32}N\rho_L. \quad (26)$$

The choices were made in the legal order

$$\text{fixed real parameters, } \lambda(k) \longrightarrow K_b \longrightarrow L \longrightarrow K_o \longrightarrow N.$$

Note that λ depends only on k , so it may be fixed at the very start; the cutoff L then depends on k through λ , C_k and $\tau(k)$, and N comes last. There is no circularity.

Let $\Gamma_N = \{a \in O_N : \exists b \in B_N, a + \lambda b \mid k\lambda ab\}$; since $B_N \subseteq S_N$, $|\Gamma_N| \leq E_N$. Put

$$A_N = (O_N \setminus \Gamma_N) \cup \{\lambda b : b \in B_N\} \subseteq \{1, \dots, \lambda N\},$$

a disjoint union because λ is even. Two odd members do not conflict, since their sum is even while kab is odd. Two even members $\lambda b, \lambda d$ conflict exactly when $b + d \mid k\lambda bd$, excluded by the $k\lambda$ -admissibility of B_N and Proposition 2.4. Every mixed conflict was removed with Γ_N . Hence A_N is k -admissible, and by (24), (25) and (26),

$$|A_N| = |O_N| - |\Gamma_N| + |B_N| \geq \frac{\lambda N}{2} - \frac{1}{32}N\rho_L - \frac{1}{32}N\rho_L + \frac{1}{8}N\rho_L = \frac{\lambda N}{2} + \frac{1}{16}N\rho_L.$$

The selected L is an absolute constant depending only on k , so $\eta_k = \rho_L/16 > 0$. $\square$

Proof of Theorem 1.2. Apply Theorem 6.2 with $N = \lfloor X/\lambda \rfloor$. For all large X the resulting set lies in $\{1, \dots, X\}$ and has at least

$$\frac{\lambda}{2} \lfloor X/\lambda \rfloor + \eta_k \lfloor X/\lambda \rfloor \geq \frac{X - \lambda}{2} + \frac{\eta_k}{2\lambda} X \geq \left(\frac{1}{2} + \frac{\eta_k}{4\lambda}\right) X$$

elements, the last step holding for X large enough that $\frac{\eta_k}{4\lambda} X \geq \frac{\lambda}{2}$. So $\varepsilon_k = \eta_k/(4\lambda) = \rho_L/(64\lambda)$ works. $\square$

7 Even multipliers: what is true and what obstructs

For even k the only conclusion we obtain is Theorem 1.3. This section explains why, and records the structural facts that any future attempt will have to confront.

7.1 No dense starting layer

The two-layer construction of Section 6 needs a set of density $1/2 - o(1)$ that is k -admissible *before* any deletion, because the whole gain η_k is of size ρ_L , far smaller than any loss proportional to the ambient density. For odd k the odd numbers supply that layer. By Theorem 1.1 no periodic set supplies it when k is even, and by Proposition 2.4 passing to a dilated layer λA replaces k by $k\lambda$, which is again even. There is therefore no scale at which a periodic layer becomes available.

7.2 Roughness caps the source

The output of Section 4 has density $\asymp \rho_L$, and this is not an artefact of the write-up. What (8) bounds is the bad *proportion* $T/|S|$, and it bounds it by $\ll 4^K (\log L)^{-3/2}$; any negative power of $\log L$ would do, and the exponent $-3/2$ has no significance for closing the argument. What matters is that the estimate is useful only when $\log L$ is large, since 4^K must first be absorbed and $K \geq K_0$ is forced by Lemma 4.1. So L must be taken large — and the source S itself has density $\frac{1-\delta}{2}\rho_L$, which tends to 0 with L . The method buys a small bad proportion by paying a small absolute density, and the two cannot be decoupled: at bounded L the estimate is vacuous. This is why the method delivers positive density and nothing approaching $1/2$.

7.3 The two-adic splitting

Corollary 2.2 does give genuine structure for even k . For $k = 2$ the conflict graph on the odd numbers is the disjoint union of its restrictions to the classes 1 and 3 modulo 4. This reduces the problem to a single class of density $1/4$, but does not solve it: both classes contain conflicts, the first being $(5, 45)$ and $(3, 15)$ respectively, and inside a class the criterion of Lemma 2.1 becomes $\frac{u+v}{2} \mid g$, structurally the same divisibility problem one started with.

7.4 A heuristic, not a theorem

Take $k = 2$. For coprime odd u, v with $u \equiv v \pmod{4}$, write $y = (u + v)/2$, which is odd by Corollary 2.2. By Lemma 2.1 the pair

$$a = tyu, \quad b = tyv \quad (t \text{ odd})$$

is a 2-conflict, and $\Omega(a) \leq \Omega(b)$ exactly when $\Omega(u) \leq \Omega(v)$; in that case the orientation deletes b . Taking $u = 1$ already shows that every odd multiple of $v(v + 1)/2$ with $v \equiv 1 \pmod{4}$ is deleted, and summing the densities $\asymp 1/(v(u + v))$ of all these families over the pairs with $\Omega(u) \leq \Omega(v)$ gives a divergent total, since for a typical v the inner sum $\sum_u u^{-1} \mathbf{1}_{\Omega(u) \leq \Omega(v)}$ is unbounded. This suggests that almost every odd number is the Ω -heavy endpoint of some 2-conflict, so that Sawin's orientation cannot retain density $1/2 - o(1)$ from the odd numbers. We emphasise that this is a heuristic about the *method*: a divergent first moment is not a proof that the orientation deletes almost everything, and it says nothing at all about f_2 itself, since a graph with minimum degree at least one may still have an independent set of density $1 - o(1)$. We have not proved either statement.

7.5 The largest-prime-factor fibration

The thread contains an observation of StijnC: the set of $n \leq N$ with $P^+(n) \geq \sqrt{kN}$ has density $\log 2 \approx 0.693$ and, when the largest prime factors differ, contains no conflict. The argument as posted leaves the equal-prime case open. It can be completed, and the completed statement explains why the observation cannot be used to beat $1/2$.

Proposition 7.1 (fibration). *Let $k \geq 1$, $N \geq 1$ and $P_0 > \sqrt{\max(k, 2)N}$. For each prime $p \in (P_0, N]$ let $A_p \subseteq \{1, \dots, \lfloor N/p \rfloor\}$ be k -admissible. Then $\bigcup_{P_0 < p \leq N} pA_p$ is a k -admissible subset of $\{1, \dots, N\}$ of size $\sum_p |A_p|$. Consequently*

$$f_k(N) \geq \sum_{P_0 < p \leq N} f_k(\lfloor N/p \rfloor).$$

Proof. No $n \leq N$ is divisible by two primes exceeding $P_0 > \sqrt{N}$, so the union is disjoint and each element determines its fibre.

Let $a = pm_1$ and $b = qm_2$ with $p \neq q$. Then $p \nmid b$ and $q \nmid a$, since otherwise $pq \mid b \leq N < pq$. Suppose $kab = (a + b)h$. As $p \mid a$ and $p \nmid b$ we have $p \nmid a + b$, so $v_p(h) = v_p(kab) \geq 1$; likewise $v_q(h) \geq 1$; hence $pq \mid h$ and

$$kab = (a + b)h \geq pq(a + b) > \max(k, 2)N(a + b) \geq kN(a + b) > kab,$$

using $N \geq a$ and $a + b > b$. This contradiction rules out conflicts across distinct fibres.

Let $a = pm_1$ and $b = pm_2$ with $m_1 \neq m_2$. Put $g' = (m_1, m_2)$ and $x' = (m_1 + m_2)/g'$. By Lemma 2.1 applied to the pair (a, b) , whose gcd is pg' and whose reduced sum is x' , the conflict is equivalent to $x'/(x', k) \mid pg'$. Now $x' \leq m_1 + m_2 \leq 2N/p < p$ because $p > P_0 > \sqrt{2N}$, so $p \nmid x'$ and the condition reduces to $x'/(x', k) \mid g'$, i.e. to $m_1 + m_2 \mid km_1m_2$. Hence a conflict inside a fibre is exactly a conflict in A_p . $\square$

Remark 7.2. Since $\sum_{P_0 < p \leq N} 1/p = \log \frac{\log N}{\log P_0} + o(1) \rightarrow \log 2$ for $P_0 \asymp \sqrt{N}$, feeding sets of density θ into Proposition 7.1 returns a set of density $\theta \log 2 < \theta$. The fibration transfers density downwards and can never bootstrap it; in particular it gives no new lower bound for any k .

8 Flagged items

We list every point at which this manuscript relies on something it does not prove, or at which we have altered the source presentation.

1. **Dependence on the companion manuscript.** Theorem 1.2 depends on [3], which is an unrefereed proof claim. If its Theorem 1.1 is wrong, Theorem 1.2 is wrong for every odd k , including $k = 1$. Theorem 1.3 depends only on Section 4, whose method is Sawin’s published one; and Theorem 1.1, Lemma 2.1, Corollary 2.2, Proposition 2.4 and Proposition 7.1 are independent of both and are proved here in full.
2. **The two analytic inputs.** Proposition 3.1 and Proposition 3.2 are quoted, not proved. In particular the uniformity of the implied constant in (4) over the multiplicative class, which is what allows F to depend on the cutoff L , is the form in which [3] uses [2, Thm. 3.1]; we use it identically and have not independently re-derived it.
3. **Ineffectivity.** Both quoted inputs carry unspecified absolute constants, so ε_k and c_k are not effective as stated. Even granting effective inputs, the cutoff L must satisfy $(\log L)^{1-c \log q_o}$ larger than a fixed multiple of the k -dependent constant C_k of (17), and $1 - c \log q_o < 3 \cdot 10^{-4}$; the resulting ε_k is astronomically small. We make no claim about its size.
4. **The $\Omega_{\geq L}(w)$ majorisation.** In the proof of Proposition 5.3 we use $\Omega_L(w, x) \geq \Omega_{\geq L}(w) - 1$ where [3, Prop. 5.2] writes an equality. The inequality is correct and costs one factor q_o ; the equality as written is not justified for those w carrying a prime factor in $(x, 2\delta^{-1}x]$. The supplement verifies exhaustively that at most one such prime factor can occur.
5. **Second residual ranges.** The treatment of the second residual term in Proposition 4.3(iv) and in Proposition 5.3 is by reference to the corresponding passages of [3], with the observation that the multiplier only changes the relevant D and Y by bounded factors. We have checked that no exponent changes, but we have not rewritten those two dyadic splittings in full here.
6. **The dilation.** An earlier draft of this manuscript used the dilation 2 throughout, copying [3]. That is *false* for every odd k divisible by 3 or 5: Remark 5.1 exhibits the offending families, which in the coordinates of Lemma 5.2 are exactly those with $u = 1$, and which carry a positive proportion of the source. The error was found by an adversarial audit of the draft. The fix is (12), and it is the reason Proposition 5.3 now carries the hypothesis $L > \lambda(1 + \delta^{-1})$ and the conclusion (15). We record this because the corresponding step in [3] — the assertion that every relevant dyadic block has $X \gg L$ — is stated there without proof; it is true at $k = 1$, for the reason given in the proof of (15), but it is not automatic.
7. **Triviality for odd k .** Theorem 1.3 has content only for even k ; for odd k the odd numbers already give density $1/2$. We state it uniformly for convenience, not because the odd case is a result.
8. **Even k .** We do not improve Sawin’s constant at $k = 2$, we do not answer the thread question $f_2(N) \geq (1/2 + o(1))N$, and we give no explicit c_k for any $k \geq 2$.

9. **The heuristic of Section 7.** Explicitly not a theorem, as stated there.
10. **No formalization.** A Lean 4 development for this manuscript is planned and has not been written. The companion manuscript’s formalization covers $k = 1$ and $k = 2$ only and does not certify any statement here.

9 Verification, prior work, and disclosure

9.1 The exact verifier

The supplement `paper/supplement/verify_certificate.py` is a standalone standard-library Python program using exact integer and rational arithmetic; no floating-point comparison is load-bearing. It certifies, in ten groups: the universal periodic witness (2) and the resulting benchmark densities of Theorem 1.1; the conflict criterion of Lemma 2.1; the two-adic restriction of Corollary 2.2 and the mod-4 splitting at $k = 2$; the generalised source coordinates of Lemma 4.2, including $u \geq L$, $x > L$ and the exact L -smooth part; the mixed coordinates of Lemma 5.2 and their converse, the exclusion (15) of the $u = 1$ families for the repaired dilation together with a negative control exhibiting those families for the dilation 2, and the at-most-one-large-prime fact used in flagged item 4; the root counts $3p - 2$ for both linear-form triples and the $p = 2$ factor; the complete small- and large-prime Euler factors for both triples, including the modified factors at primes dividing the multiplier; the five window inequalities of Lemma 6.1 with rational logarithm intervals of width below 10^{-80} , together with the symbolic cancellation of the $\log L$ exponents to $-3/2$ in the source assembly and -2 in the mixed assembly, carried out as an identity in the formal symbols M, α, β, s ; and Proposition 7.1 with its contraction constant; and, finally, that the assembled two-layer set of Theorem 6.2 really is k -admissible in a finite box. The recorded output is `paper/supplement/certificate-output.txt`.

The last group deserves a caveat. It confirms the combinatorics of the assembly — that the three conflict types really are all excluded — but it cannot establish the density gain, which is asymptotic and requires L and N to be chosen in the legal order of Theorem 6.2. The boxes reported there do happen to show a positive surplus over the odd numbers, but that is a property of those particular boxes and proves nothing: no finite computation can test the inequality of Theorem 1.2. What the group does test, and what it was added to test, is that the assembled set is genuinely k -admissible — the property that the dilation 2 silently violates for $3 \mid k$ and $5 \mid k$.

These checks certify arithmetic, local densities and the numerical window. They are not, and are not offered as, evidence for the analytic inputs of Section 3.

9.2 Prior-work search

A search completed on 29 July 2026 covering arXiv, the Problem 327 page and its discussion thread, and general web search found: Sawin’s paper [7], which treats $k = 2$ only; the upper bounds of [6] for $k = 1, 2, 3$; Sawin’s thread comment giving the greedy lower bound $N/(\log N)^{0.3588\dots+o(1)}$ for $k = 2$; and no lower bound for f_k with $k \geq 3$ other than the odd numbers for odd k , and in particular none at all for even $k \geq 4$. We located no prior treatment of the periodic benchmark of Theorem 1.1. This is a dated search report, not a categorical priority claim.

9.3 Computational and AI-use disclosure

AI systems were used for literature search, symbolic and numerical exploration, adversarial auditing of the generalisation, independent reconstruction of intermediate estimates, and drafting. The

identification of the periodic benchmark, the choice of what does and does not transfer from [3], and the decision to report Theorem 1.2 and Theorem 1.3 rather than a stronger claim are the author's. The author is responsible for all claims and errors.

References

- [1] Thomas F. Bloom. Erdos Problem 327. <https://www.erdosproblems.com/327>. Accessed 29 July 2026.
- [2] Régis de la Bretèche and Gérald Tenenbaum. Mean values of arithmetic functions and application to sums of powers. *Mathematical Proceedings of the Cambridge Philosophical Society*, 180, 2026. Originally arXiv:2403.19320.
- [3] Donald Della Pietra. A positive-density improvement over the odd numbers in Erdős problem 327. <https://github.com/donalddellapietra/erdos-327-proof>, 2026. Unrefereed preprint and proof claim; the manuscript is `paper/main.tex` in the same repository as the present one.
- [4] Erdős Problems contributors. 327 Discussion Thread. <https://www.erdosproblems.com/forum/discuss/327>. Accessed 29 July 2026.
- [5] Paul Erdos and Ronald L. Graham. *Old and New Problems and Results in Combinatorial Number Theory*, volume 28 of *Monographies de L'Enseignement Mathématique*. L'Enseignement Mathématique, Geneva, 1980.
- [6] leon2k2k2k. Upper bounds for Erdős problem 327. <https://leon2k2k2k.github.io/erdos327.pdf>, 2026. Posted in the Problem 327 discussion thread, 13 May 2026; independently checked by Nat Sothanaphan.
- [7] Will Sawin. Sets of unit fractions without two members whose average is a unit fraction. arXiv:2607.15419 [math.NT], 2026.
- [8] Gérald Tenenbaum. *Introduction to Analytic and Probabilistic Number Theory*, volume 163 of *Graduate Studies in Mathematics*. American Mathematical Society, Providence, RI, third edition, 2015.